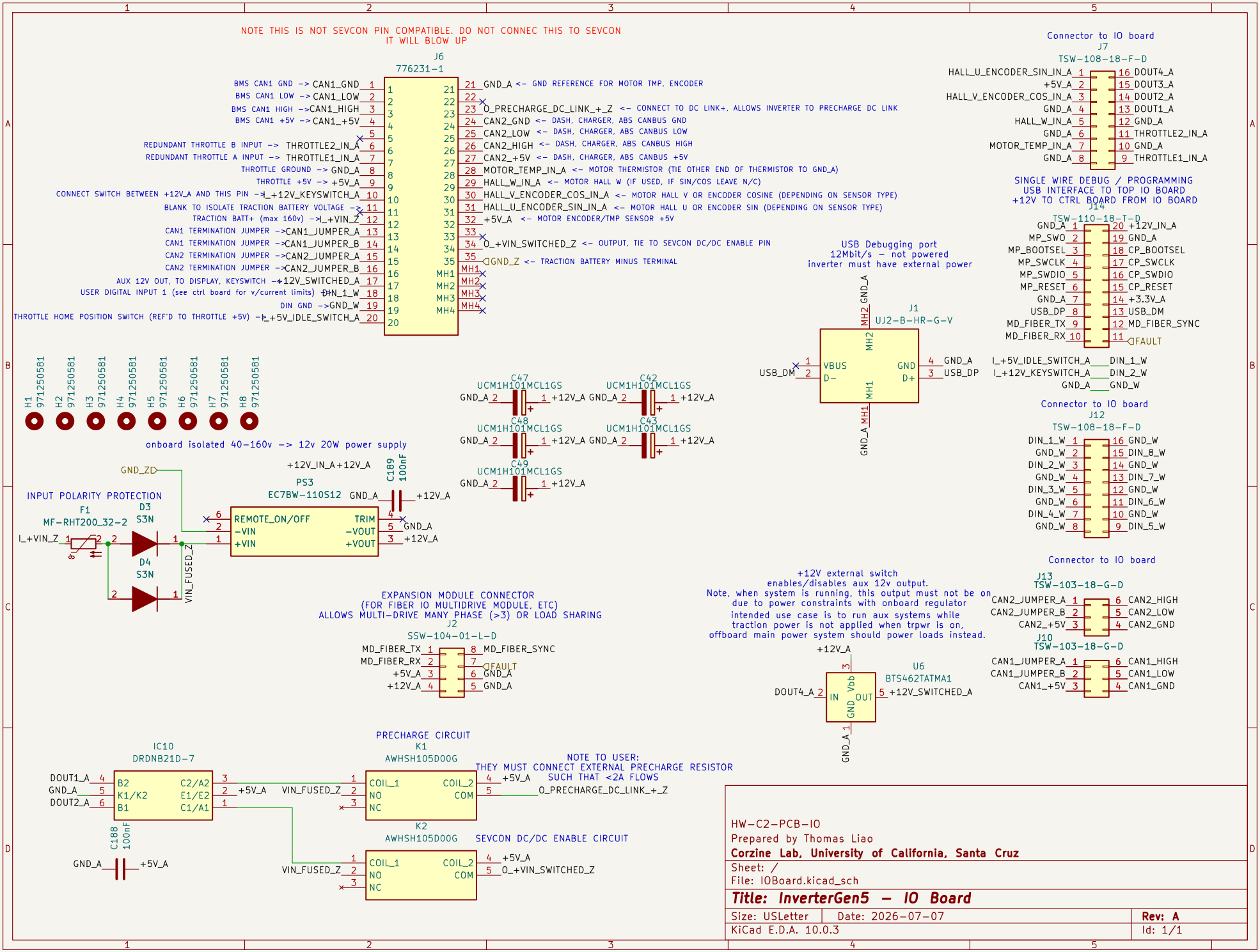
NOTE THIS IS NOT SEVCON PIN COMPATIBLE. DO NOT CONNEC THIS TO SEVCON
IT WILL BLOW UP
J6
776231-1
BMS CAN1 GND -> CAN1_GND 1
BMS CAN1 LOW -> CAN1_LOW 2
BMS CAN1 HIGH -> CAN1_HIGH 3
BMS CAN1 +5V -> CAN1_+5V 4
REDUNDANT THROTTLE B INPUT -> THROTTLE2_IN_A 6
REDUNDANT THROTTLE A INPUT -> THROTTLE1_IN_A 7
THROTTLE GROUND -> GND_A 8
THROTTLE +5V -> +5V_A 9
CONNECT SWITCH BETWEEN +12V_A AND THIS PIN -> I_+12V_KEYSWITCH_A 10
BLANK TO ISOLATE TRACTION BATTERY VOLTAGE -> 11
TRACTION BATT+ (max 160v) -> I_+VIN_Z 12
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_A 13
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_B 14
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_A 15
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_B 16
AUX 12V OUT, TO DISPLAY, KEYSWITCH -> +12V_SWITCHED_A 17
USER DIGITAL INPUT 1 (see ctrl board for v/current limits) -> DIN_1_W 18
DIN GND -> GND_W 19
THROTTLE HOME POSITION SWITCH (REF'D TO THROTTLE +5V) -> I_+5V_IDLE_SWITCH_A 20
21 GND_A <- GND REFERENCE FOR MOTOR TMP, ENCODER
23 O_PRECHARGE_DC_LINK_+_Z <- CONNECT TO DC LINK+, ALLOWS INVERTER TO PRECHARGE DC LINK
24 CAN2_GND <- DASH, CHARGER, ABS CANBUS GND
25 CAN2_LOW <- DASH, CHARGER, ABS CANBUS LOW
26 CAN2_HIGH <- DASH, CHARGER, ABS CANBUS HIGH
27 CAN2_+5V <- DASH, CHARGER, ABS CANBUS +5V
28 MOTOR_TEMP_IN_A <- MOTOR THERMISTOR (TIE OTHER END OF THERMISTOR TO GND_A)
29 HALL_W_IN_A <- MOTOR HALL W (IF USED, IF SIN/COS LEAVE N/C)
30 HALL_V_ENCODER_COS_IN_A <- MOTOR HALL V OR ENCODER COSINE (DEPENDING ON SENSOR TYPE)
31 HALL_U_ENCODER_SIN_IN_A <- MOTOR HALL U OR ENCODER SIN (DEPENDING ON SENSOR TYPE)
32 +5V_A <- MOTOR ENCODER/TMP SENSOR +5V
34 O_+VIN_SWITCHED_Z <- OUTPUT, TIE TO SEVCON DC/DC ENABLE PIN
35 GND_Z <- TRACTION BATTERY MINUS TERMINAL
MH1 MH2 MH3 MH4
Connector to IO board
J7
TSW-108-18-F-D
HALL_U_ENCODER_SIN_IN_A 1
+5V_A 2
HALL_V_ENCODER_COS_IN_A 3
GND_A 4
HALL_W_IN_A 5
GND_A 6
MOTOR_TEMP_IN_A 7
GND_A 8
16 DOUT4_A
15 DOUT3_A
14 DOUT2_A
13 DOUT1_A
12 GND_A
11 THROTTLE2_IN_A
10 GND_A
9 THROTTLE1_IN_A
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
TSW-110-18-T-D
GND_A 1
MP_SWO 2
MP_BOOTSEL 3
MP_SWCLK 4
MP_SWDIO 5
MP_RESET 6
GND_A 7
USB_DP 8
MD_FIBER_TX 9
MD_FIBER_RX 10
20 +12V_IN_A
19 GND_A
18 CP_BOOTSEL
17 CP_SWCLK
16 CP_SWDIO
15 CP_RESET
14 +3.3V_A
13 USB_DM
12 MD_FIBER_SYNC
11 FAULT
I_+5V_IDLE_SWITCH_A DIN_1_W
I_+12V_KEYSWITCH_A DIN_2_W
GND_A GND_W
USB Debugging port
12Mbit/s - not powered
inverter must have external power
J1
UJ2-B-HR-G-V
MH2 GND_A
MH2
VBUS 1
D- 2
GND 4 GND_A
D+ 3 USB_DP
MH1
GND_A MH1
USB_DM
H1 971250581
H2 971250581
H3 971250581
H4 971250581
H5 971250581
H6 971250581
H7 971250581
H8 971250581
C47
UCM1H101MCL1GS
GND_A 2
1 +12V_A
C48
UCM1H101MCL1GS
GND_A 2
1 +12V_A
C49
UCM1H101MCL1GS
GND_A 2
1 +12V_A
C42
UCM1H101MCL1GS
GND_A 2
1 +12V_A
C43
UCM1H101MCL1GS
GND_A 2
1 +12V_A
Connector to IO board
J12
TSW-108-18-F-D
DIN_1_W 1
GND_W 2
DIN_2_W 3
GND_W 4
DIN_3_W 5
GND_W 6
DIN_4_W 7
GND_W 8
16 GND_W
15 DIN_8_W
14 GND_W
13 DIN_7_W
12 GND_W
11 DIN_6_W
10 GND_W
9 DIN_5_W
onboard isolated 40-160v -> 12v 20W power supply
GND_Z
INPUT POLARITY PROTECTION
F1
MF-RHT200_32-2
I_+VIN_Z 1
D3
S3N
D4
S3N
VIN_FUSED_Z
+12V_IN_A +12V_A
PS3
EC7BW-110S12
REMOTE_ON/OFF 6
-VIN 2
+VIN 1
TRIM 4
-VOUT 5 GND_A
+VOUT 3 +12V_A
C189
100nF
GND_A
+12V_A
EXPANSION MODULE CONNECTOR
(FOR FIBER IO MULTIDRIVE MODULE, ETC)
ALLOWS MULTI-DRIVE MANY PHASE (>3) OR LOAD SHARING
J2
SSW-104-01-L-D
MD_FIBER_TX 1
MD_FIBER_RX 2
+5V_A 3
+12V_A 4
8 MD_FIBER_SYNC
7 FAULT
6 GND_A
5 GND_A
+12V external switch
enables/disables aux 12v output.
Note, when system is running, this output must not be on
due to power constraints with onboard regulator
intended use case is to run aux systems while
traction power is not applied when trpwr is on,
offboard main power system should power loads instead.
+12V_A
U6
BTS462TATMA1
DOUT4_A 2
IN
Vbb 3
GND
OUT 5 +12V_SWITCHED_A
GND_A 1
Connector to IO board
J13
TSW-103-18-G-D
CAN2_JUMPER_A 1
CAN2_JUMPER_B 2
CAN2_+5V 3
6 CAN2_HIGH
5 CAN2_LOW
4 CAN2_GND
J10
TSW-103-18-G-D
CAN1_JUMPER_A 1
CAN1_JUMPER_B 2
CAN1_+5V 3
6 CAN1_HIGH
5 CAN1_LOW
4 CAN1_GND
IC10
DRDNB21D-7
DOUT1_A 4
GND_A 5
DOUT2_A 6
B2
K1/K2
B1
C2/A2 3
E1/E2 2 +5V_A
C1/A1 1
C188
100nF
GND_A
+5V_A
PRECHARGE CIRCUIT
K1
AWHSH105D00G
VIN_FUSED_Z 1
NO 2
NC 3
COIL_1
COIL_2
COM
4 +5V_A
5 O_PRECHARGE_DC_LINK_+_Z
NOTE TO USER:
THEY MUST CONNECT EXTERNAL PRECHARGE RESISTOR
SUCH THAT <2A FLOWS
K2
AWHSH105D00G
SEVCON DC/DC ENABLE CIRCUIT
VIN_FUSED_Z 1
NO 2
NC 3
COIL_1
COIL_2
COM
4 +5V_A
5 O_+VIN_SWITCHED_Z
HW-C2-PCB-IO
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: IOBoard.kicad_sch
Title: InverterGen5 - IO Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1